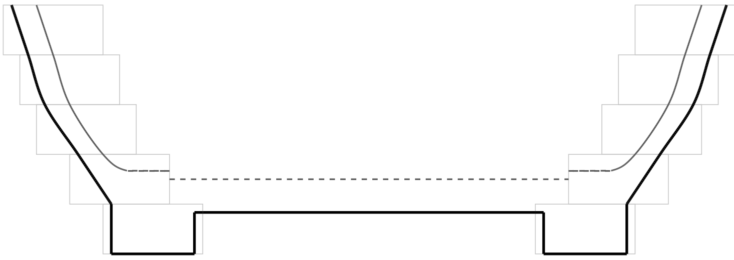
Medium Bowl Shop Plan

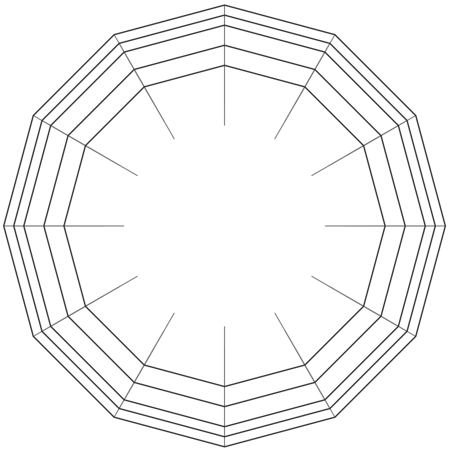
279.4 mm across · 95.3 mm tall · 5 rings · Walnut, Maple



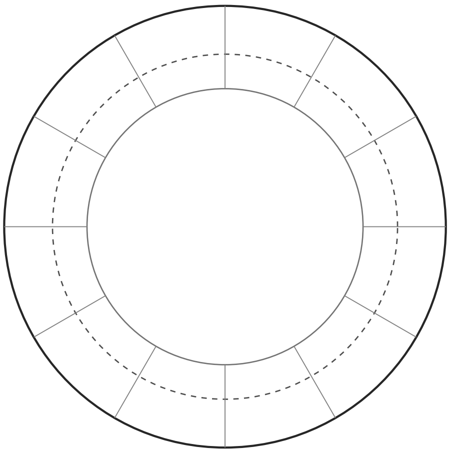
Height 95.3 mm

Max diameter 279.4 mm

Solid: turned outside. Dashed: turned inside. Faint: ring stack before turning.



Plan - from above



Bottom - foot ring and disk

Scan to customize this design

Medium Bowl Shop Plan

Project

Overall height	95.3 mm
Maximum outside diameter	279.4 mm
Ring count	5
Bottom style	Rabbeted
Target wall thickness	9.5 mm
Turning allowance	3.2 mm
Minimum overlap	12.7 mm
Expert Mode	Off
Species	Walnut, Maple

Ring numbering

Ring 1 is the foot. Rings are numbered from the bottom.

Cut list

Rings are listed bottom to top. Ring 1 is the foot.

Ring 1 Foot ring · Walnut
12 segments · thickness 19.0 mm · radial width 38.1 mm
OD 203.2 mm · ID 127.0 mm · miter 15° each end
Outside edge 54.4 mm · inside edge 34.0 mm (verify) · dividers - · make 12 · bare stock 706.3 mm

Ring 2 Ordinary segmented ring · Maple
12 segments · thickness 19.0 mm · radial width 38.1 mm
OD 228.6 mm · ID 152.4 mm · miter 15° each end
Outside edge 61.3 mm · inside edge 40.8 mm (verify) · dividers - · make 12 · bare stock 788.0 mm

Ring 3 Ordinary segmented ring · Walnut
12 segments · thickness 19.0 mm · radial width 38.1 mm
OD 254.0 mm · ID 177.8 mm · miter 15° each end
Outside edge 68.1 mm · inside edge 47.6 mm (verify) · dividers - · make 12 · bare stock 869.6 mm

Ring 4 Ordinary segmented ring · Maple
12 segments · thickness 19.0 mm · radial width 38.1 mm
OD 266.7 mm · ID 190.5 mm · miter 15° each end
Outside edge 71.5 mm · inside edge 51.0 mm (verify) · dividers - · make 12 · bare stock 910.5 mm

Ring 5 Ordinary segmented ring · Walnut
12 segments · thickness 19.0 mm · radial width 38.1 mm
OD 279.4 mm · ID 203.2 mm · miter 15° each end
Outside edge 74.9 mm · inside edge 54.4 mm (verify) · dividers - · make 12 · bare stock 951.3 mm

Materials

Walnut - clear finished strips, 19.0 mm thick x 38.1 mm wide

PLAN TO YIELD

3.00 metres of clear, finished strip

2368.5 mm segments + 118.3 mm saw kerf (36 cuts) + 13.5 mm setup = 2500.3 mm
+ 10% planning allowance = 2750.3 mm, rounded up

- Ring 1 - 12 segments at 54.4 mm
- Ring 3 - 12 segments at 68.1 mm
- Ring 5 - 12 segments at 74.9 mm

Maple - clear finished strips, 19.0 mm thick x 38.1 mm wide

PLAN TO YIELD

2.00 metres of clear, finished strip

1592.6 mm segments + 78.9 mm saw kerf (24 cuts) + 13.5 mm setup = 1685.0 mm
+ 10% planning allowance = 1853.5 mm, rounded up

- Ring 2 - 12 segments at 61.3 mm
- Ring 4 - 12 segments at 71.5 mm

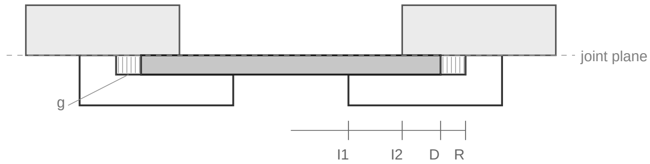
The 10% planning allowance is a contingency, not a yield guarantee. Select enough rough stock to yield these clear, finished strips after milling, ripping, end trimming and defect removal. A wide board may yield several strips; a defective one may yield far less. The allowance does not cover knots, checks, sapwood or other mapped defects. Compatible rings may share a strip.

Bottom construction

(I1)	Ring 1 inside diameter	**127.0 mm**
(I2)	Ring 2 inside diameter	**152.4 mm**
(D)	Disk diameter	**158.8 mm**
(R)	Rabbet recess diameter	**161.9 mm**
(T)	Disk thickness	**9.5 mm**
(B)	Rabbet depth	**9.5 mm**
(g)	Radial expansion clearance	**1.6 mm per side**
	Support overlap on ring 1	**15.9 mm per side**
(c)	Capture overlap under ring 2	**3.2 mm per side**
(F)	Panel blank - square	**178.8 mm**
	Disk species	**Walnut**

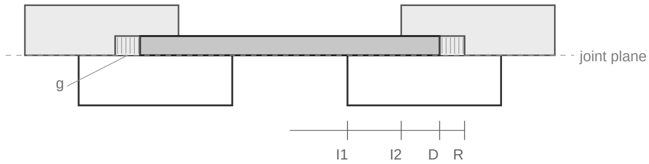
I2 < D < R. The disk reaches under ring 2 by 3.2 mm; the clearance lies beyond the disk, between its edge and the recess wall.

Rabbet Ring 1 from above

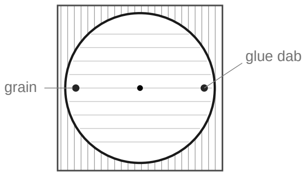


Interior floor at the joint plane.
Not to scale.

Rabbet Ring 2 from below



Interior floor one disk thickness (9.5 mm) above the joint plane.
Not to scale.



Panel blank - not to scale
Dabs sit on the face, inside the captured rim.

Notes

The disk size is governed by the larger opening in ring 2. It must extend beneath ring 2 far enough to be captured, regardless of which ring receives the rabbet.

Both constructions therefore use the same disk diameter and rabbet recess diameter. Ring 1 supplies the lower support; ring 2 supplies the upper capture.

Rabbet ring 1 from above to place the finished interior floor at the ring joint. Rabbet ring 2 from below to raise the finished interior floor by one disk thickness. Choose the location that preserves adequate ring thickness and suits the intended interior profile.

Make the rabbet depth equal to the finished disk thickness when a flush fit is required. Fit the disk with the specified radial expansion clearance and do not glue it rigidly around its circumference.

To keep the disk from sliding freely in the expansion clearance, it is acceptable to place one small glue dab on the FACE of the disk within the captured rim, at either end of the grain axis - where the longest grain line through the centre meets the circumference. Do not place a dab on the disk edge, and do not let glue enter the expansion gap.

The square panel blank includes a machining allowance of 10 mm per side.
It is not part of the trapping geometry.

Print at Actual Size / 100%. Do not use Fit or Scale to Fit.

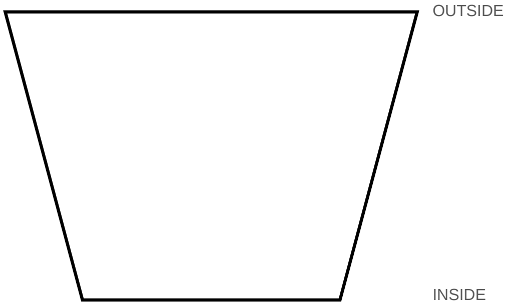
Measure both calibration references before using these templates.

CALIBRATION - measure these before using any template

1 inch

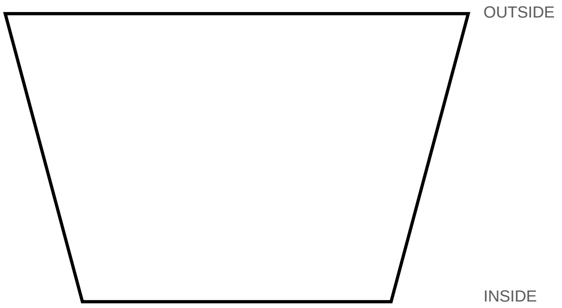
25 mm

If either reference measures wrong, these templates are invalid. Reprint at 100%.



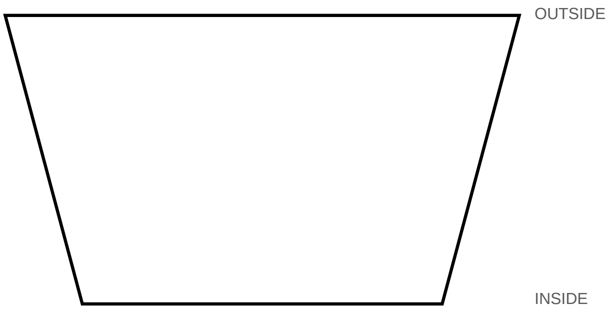
S1 · Ring 1 · make 12

12 segments · miter 15° each end
Outside 54.4 mm · Inside 34.0 mm · Width 38.1 mm
Stock thickness 19.0 mm



S2 · Ring 2 · make 12

12 segments · miter 15° each end
Outside 61.3 mm · Inside 40.8 mm · Width 38.1 mm
Stock thickness 19.0 mm



S3 · Ring 3 · make 12

12 segments · miter 15° each end
Outside 68.1 mm · Inside 47.6 mm · Width 38.1 mm
Stock thickness 19.0 mm

Print at Actual Size / 100%. Do not use Fit or Scale to Fit.

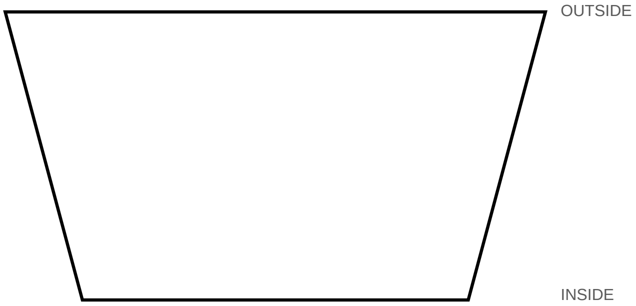
Measure both calibration references before using these templates.

CALIBRATION - measure these before using any template

1 inch

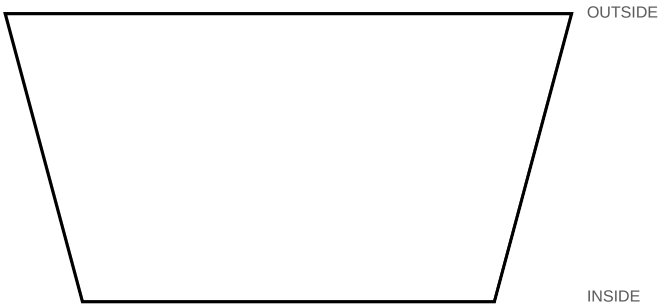
25 mm

If either reference measures wrong, these templates are invalid. Reprint at 100%.



S4 · Ring 4 · make 12

12 segments · miter 15° each end
Outside 71.5 mm · Inside 51.0 mm · Width 38.1 mm
Stock thickness 19.0 mm



S5 · Ring 5 · make 12

12 segments · miter 15° each end
Outside 74.9 mm · Inside 54.4 mm · Width 38.1 mm
Stock thickness 19.0 mm